Optimal Molecular Networks for Excitonic Energy Transport

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Outline

Light-Harvesting And The Fenna-Matthews-Olson Complex

The Surrogate Model

Metrology of Transport Efficiency

Statistics of Transport Efficiency

Optimal Configurations

Efficiency And Dephasing Noise

Light-Harvesting

The Fenna-Matthews-Olson Complex

light-harvesting absorption of a photon, transfer to reaction center

light-to-charge efficiency > 95% !

energy shuttle Fenna-Matthews-Olson (FMO) protein from green sulfur bacteria

precise functionality more or less unknown...

scope of this talk are quantum mechanical effects involved?

Conjecture

"Functional Efficiency of Biomolecular Units Stems from Quantum Coherence."

contra Why Should It Not Be The Case?

FMO complex network of nodes

coherent transport not step-by-step, but all paths simultaneously

localization disorder-induced, due to destructive interference

noise and decoherence increase conductance via transition to classical transport

optimized bath optimal control of transport

pro Why Should This Be The Case?

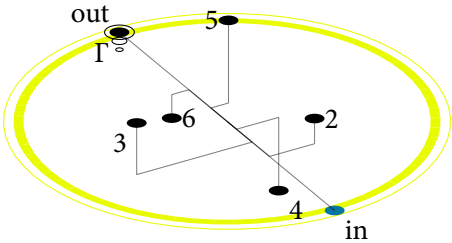
experiments confirm coherence, efficiency too high for classical explanation

constructive interference unlikely, but not impossible

optimized system nature tailors conformations with optimal *coherent* transport properties

Physical Abstraction

Coherent Dynamics on A Fully Connected, Finite Graph



couplings $V_{i,j} = V^*_{j,i} = \cancel{\mathcal{A}_{i,j}}\, r^{-3}_{i,j}$

ground state $|0\rangle$

start $|\psi(0)\rangle = |\mathrm{in}\rangle$

intermediates $\{|i\rangle\}_{i=2,\ldots,N-1}$

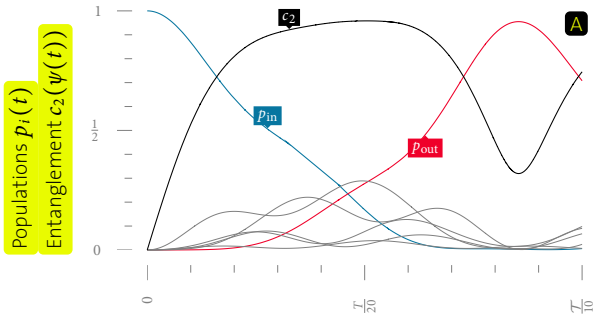
destination $|\mathrm{out}\rangle$

time scale $T = \pi/(2|V_{\mathrm{in,out}}|)$

disorder intermediate nodes uniformly distributed within sphere

Maximal Output Probability

Focus on Short Times



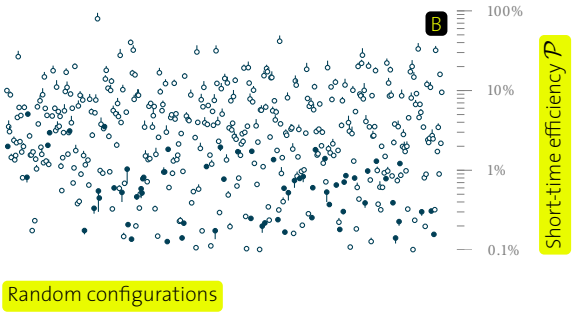


figure of merit $\mathcal{P} = \max_{t\in[0,T/10]} p_{\text{out}}(t)$

transport benefits from *rare*, constructive quantum interference

perfect transport by optimal configurations on *transient* time scales

decoherence does not change this result, $\gamma = 1/T$

Transfer Time

Focus on Long Times

sink infinite 0 K-reservoir connected to ‘out’

Poisson process Start $\longrightarrow$ Wait $\longrightarrow$ Excitation Arrived? $\xrightarrow{\text{yes}}$ Take it

no

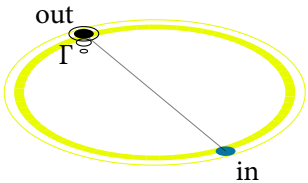
waiting time in average Γ^{-1}

description Hamiltonian $H = \cancel{\sum_{i=1}^{N} E_i \,|i\rangle\langle i|} + \sum_{i\neq j=1}^{N} V_{i,j}\,|i\rangle\langle j|$ & master equation

$$\dot{\rho}(t) = -\mathrm{i}\,[H, \rho(t)] + \Gamma\left(|0\rangle\langle \mathrm{out}|\rho(t)|\mathrm{out}\rangle\langle 0| - \tfrac{1}{2}\left\{\rho(t), |\mathrm{out}\rangle\langle \mathrm{out}|\right\}\right)$$

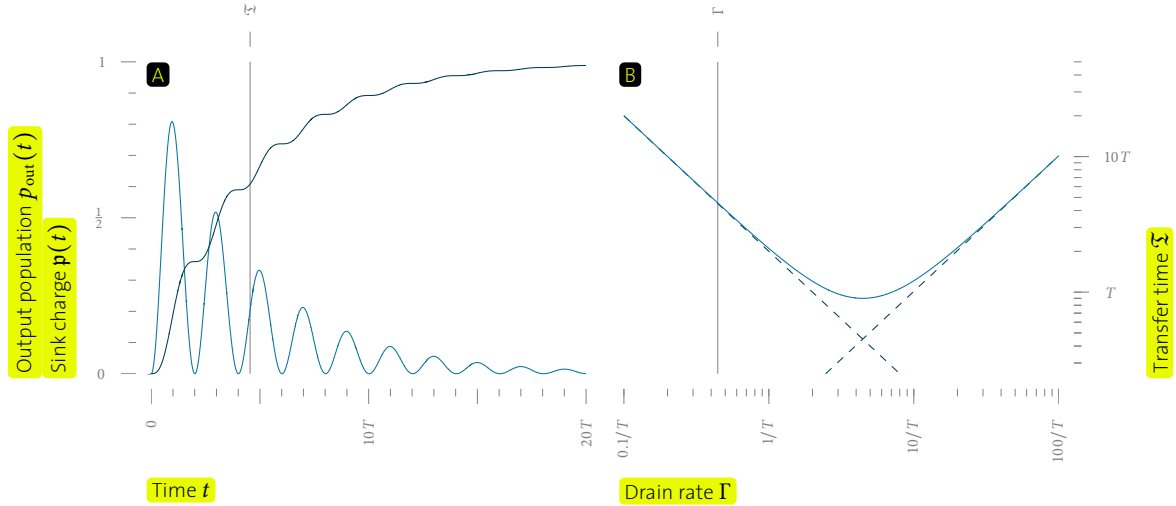
charge $\mathfrak{p}(t) = \rho_{0,0}(t) \xrightarrow{t\to\infty} 1$

transfer time $\mathfrak{T} = \int_0^\infty t\,\rho'_{0,0}(t)\,\mathrm{d}t$



Transfer Time

The Two-site Example



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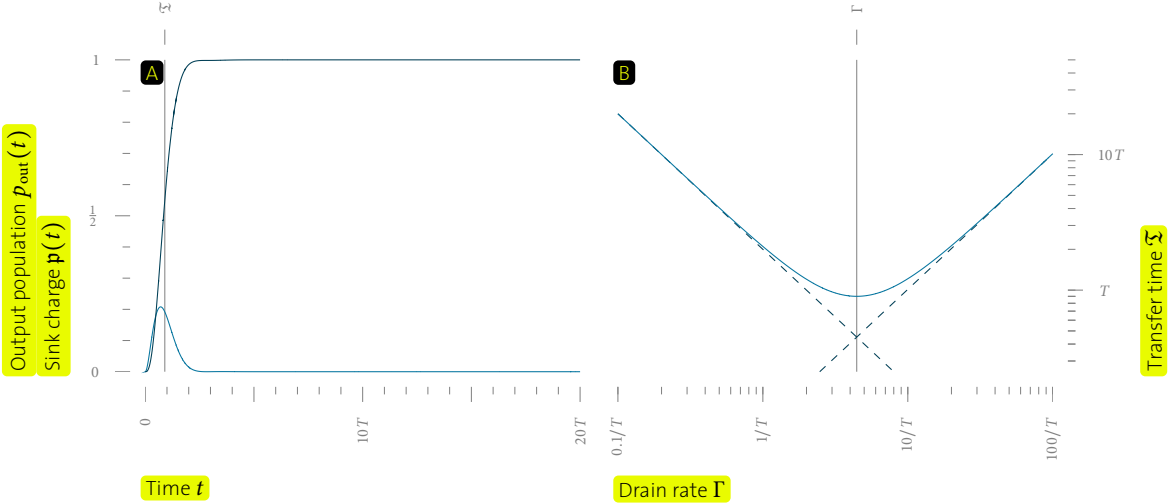

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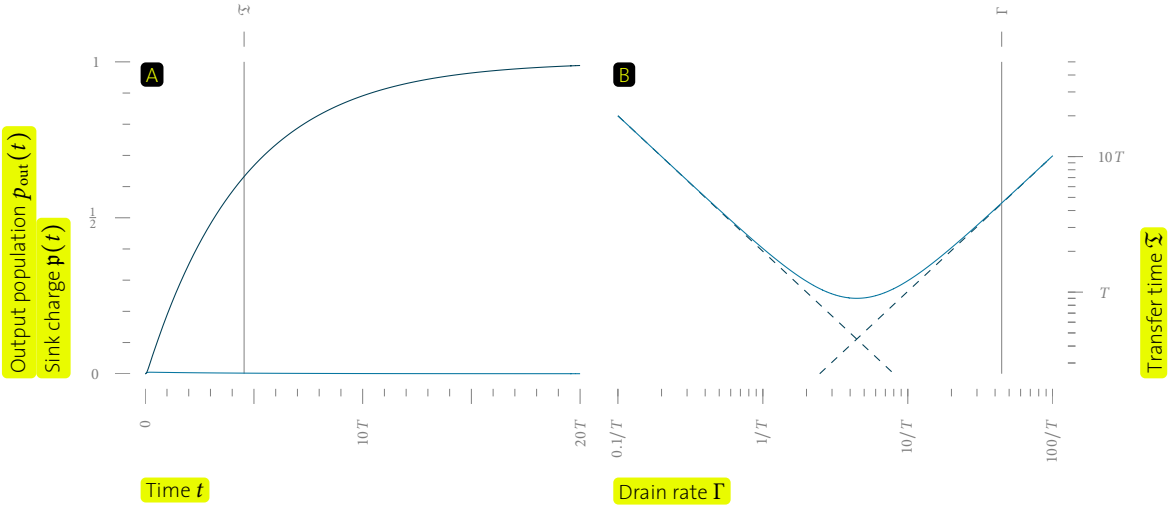
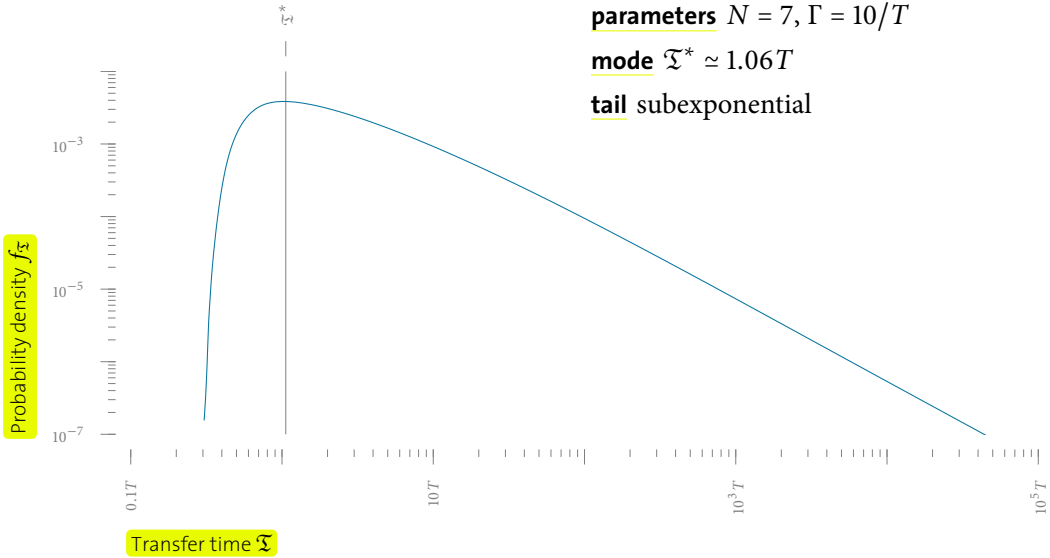


Statistics of Transport Efficiency

A Histogram of The Transfer Time



parameters $N = 7, \Gamma = 10/T$

mode $\mathfrak{T}^* \simeq 1.06T$

tail subexponential

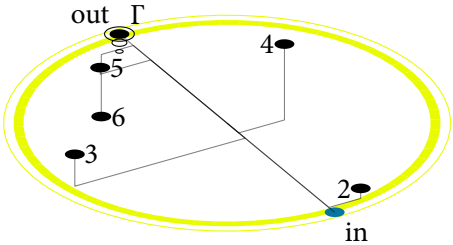
Pair Localization

blocking eigenstates $\langle \mathrm{in}|\sigma\rangle > 0$, but $\langle \mathrm{out}|\sigma\rangle = 0$

asymptotics sink eats away the rest

$$\rho(t) \sim_{t\to\infty} \left(1 - \sum_\sigma |\langle \mathrm{in}|\sigma\rangle|^2\right)|0\rangle\langle 0| + \sum_{\sigma,\sigma'} \mathrm{e}^{-\mathrm{i}(\varepsilon_\sigma - \varepsilon'_\sigma)t}|\sigma\rangle\langle\sigma|\mathrm{in}\rangle\langle\mathrm{in}|\sigma'\rangle\langle\sigma'|$$

transfer time $\mathfrak{T}$ diverges



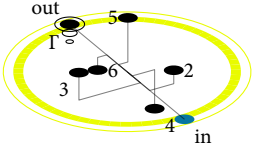
this one transfer time $\mathfrak{T} \simeq 4 \times 10^8\, T$

couples $\frac{1}{\sqrt{2}}(|\mathrm{in}\rangle \pm |2\rangle)$ approximate eigenstates

energies $\pm V \propto r_{\mathrm{in},2}^{-3}$

Minimization Algorithm

CMA-ES global, non-deterministic optimization algorithm



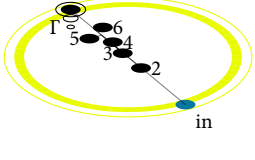
initialization random spatial arrangement

check termination criteria

sample mutations create normally distributed pool of variations

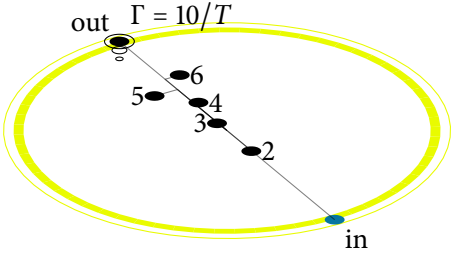
fitness competition for best $\mathfrak{T}$, select winners

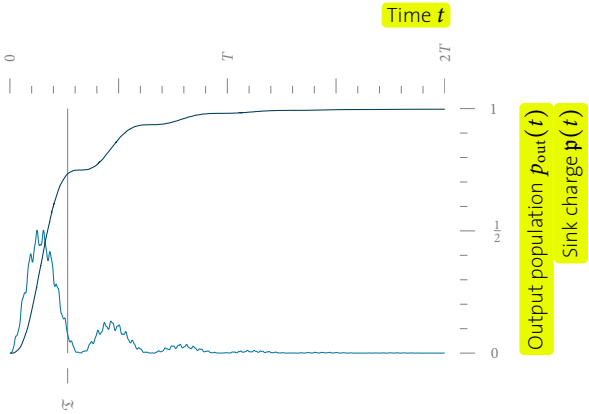
update model good attributes of winners are passed on, adjust amount of fresh diversity



stop

Optimal Solutions



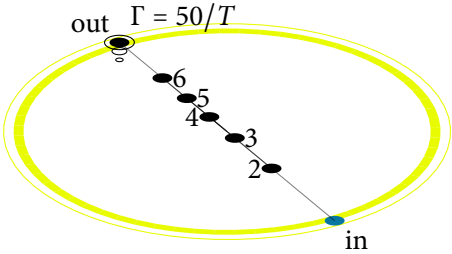


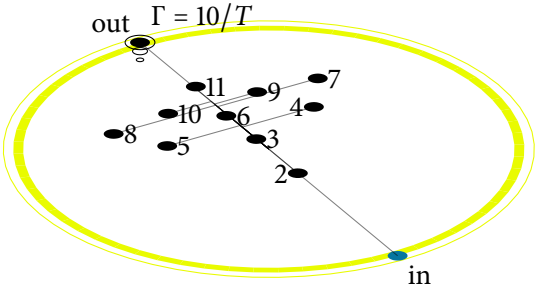
dimensionality only a few are truly 3d—many are 2d or 1d

symmetries none

stability spatial noise causes exponential degrading

Optimal Solutions

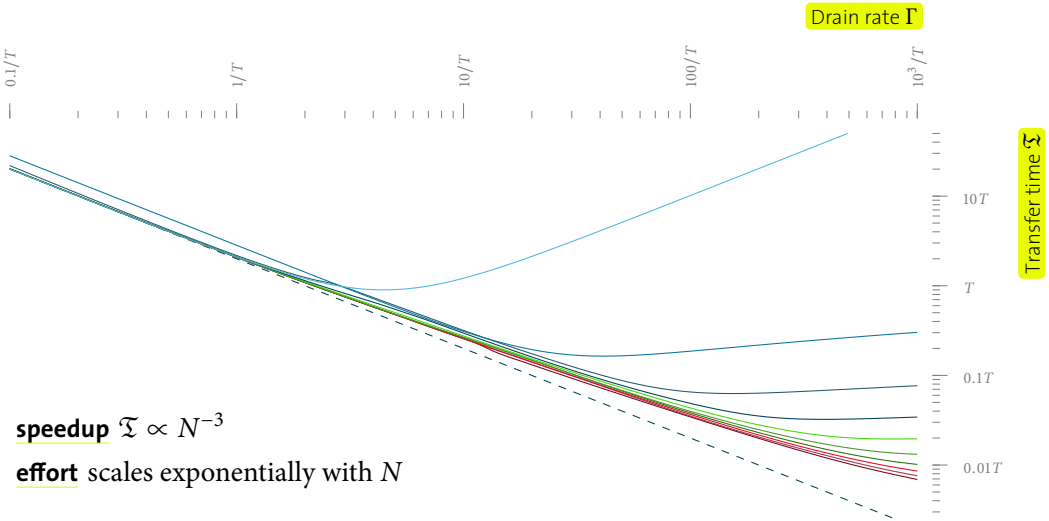




sensitivity solution depends on drain rate Γ

complexity all sites take part

The Landscape of Optimal Transport



speedup $\mathfrak{T} \propto N^{-3}$

effort scales exponentially with N

Dephasing Noise

hierarchy pyramid

peripheral ensemble dephasing

intrinsic phase diffusion

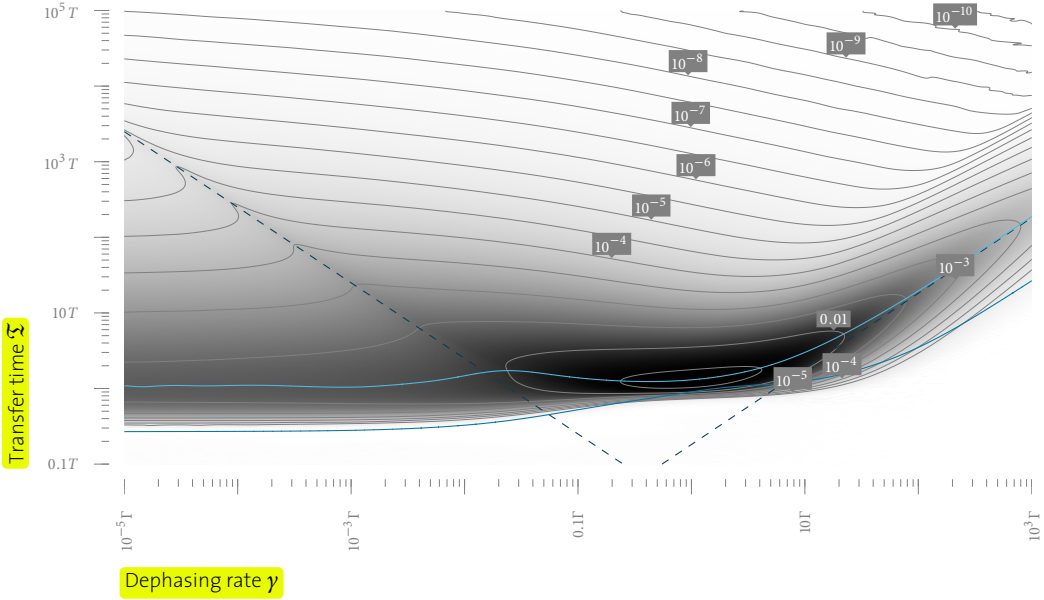
Markovian bath Lindblad-Operator

$$L_{\mathrm{deph}}(\rho) = -4\gamma \sum_{\substack{i,j=1 \\ i \neq j}}^{N} \rho_{i,j} \, |i\rangle\langle j|$$

strength homogeneous rate γ

attenuation of non-local coherences

Sweep


$10^5\,T$
$10^3\,T$
$10\,T$
$0.1\,T$
Transfer time $\mathfrak{T}$
10^{-10}
10^{-9}
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
0.01
10^{-4}
10^{-5}
$10^{-5}\,\Gamma$
$10^{-3}\,\Gamma$
0.1Γ
10Γ
$10^3\,\Gamma$
Dephasing rate γ

Summary

the optimal dephasing rate is zero If a configuration is very efficient in the noiseless situation, its performance will deplete when subjected to dephasing!

coherence is better than noise We discovered special configurations that perform more than twice as efficient than any dephasing-assisted configuration!

impact on natural evolution In a competition between “environment-assisted transport” and “quantum-interference-assisted transport” we would rather bet on the latter!